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OMDAZZ MBSL32DI

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OMDAZZ MBSL32DI 32-Channel Digital Input Isolated Acquisition Module

User Manual

1. Introduction

The MBSL32DI module is a 32-channel digital isolation input module designed for industrial applications. It communicates via the RS485 interface using the standard Modbus RTU protocol, allowing direct integration with various host computer configuration software, PLCs, and DCS systems. The module's power supply adheres to industrial-grade design standards, effectively suppressing serial mode and common mode interference to ensure stable and reliable operation.

Key features include:

- RS485 MODBUS RTU standard communication, compatible with host computer configuration software, PLCs, and industrial touch screens.
- Equipped with communication status indicators (RUN, RX, TX).
- Industrial standard compliance with isolated signal control, power supply, and RS485 communication electrical signals.
- Communication circuit features lightning protection, anti-interference design, and power polarity protection.
- RS485 communication signal output interface includes dual protection against overvoltage and overcurrent.
- Automotive-standard hardware and software dual watchdog timers ensure crash-free, stable, and reliable operation.
- Output disconnection protection: In case of a 485 network interruption, if the module does not receive data from the master station within a set time, it will automatically disconnect the output to protect field equipment.
- Communication format is software configurable.

2. SPECIFICATIONS

Item	Technical Specifications
Signal input/output	1) Input channel: 32 digital inputs
Communication output	1) Communication protocol: MODBUS-RTU 2) Interface type: isolated RS485 communication, output interface adopts double protection of overvoltage and overcurrent 3) Baud rate: 4800bps, 9600bps, 19200bps, 38400bps, 57600bps, 115200 bps 4) Parity bit: no parity, even parity, odd parity 5) Setting method: module address, baud rate, parity bit can be set by software 6) Communication distance: @9600bps 1200 meters
Module size and installation method	1.1) Installation method: standard DIN rail installation or screw installation 2) Dimensions: 125×90×43mm
Working environment	Temperature: -10□+60℃ Humidity: 35□85% (no condensation)
Power supply	1) Power supply voltage: 12V□30V wide range power supply, with power polarity protection

Figure 2.1: MBSL32DI Technical Specifications

Item	**Technical Specifications**
Model Number	MBSL32DI

Item	Technical Specifications
Type	Voltage Regulator
Signal Input/Output	32 digital inputs
Communication Protocol	MODBUS-RTU
Interface Type	Isolated RS485 communication with overvoltage and overcurrent protection
Baud Rate	4800bps, 9600bps, 19200bps, 38400bps, 57600bps, 115200 bps
Parity Bit	No parity, Even parity, Odd parity
Setting Method	Module address, baud rate, parity bit configurable by software
Communication Distance	Up to 1200 meters at 9600bps
Installation Method	Standard DIN rail installation or screw installation
Dimensions	125 × 90 × 43 mm
Operating Temperature	-10 to +60℃
Humidity	35-85% (non-condensing)
Supply Voltage	12V to 30V DC (wide range), with power polarity protection
Power Consumption	Less than 5W
Condition	New

Item	Technical Specifications
Origin	Mainland China

3. Setup and Installation

The MBSL32DI module offers flexible installation options:

- **DIN Rail Installation:** Mount the module onto a standard DIN rail for quick and secure integration into control cabinets.
- **Screw Installation:** Alternatively, the module can be securely mounted using screws through designated mounting holes.

Ensure the installation location provides adequate ventilation and maintains an operating temperature between -10℃ and +60℃, with humidity between 35% and 85% (non-condensing).

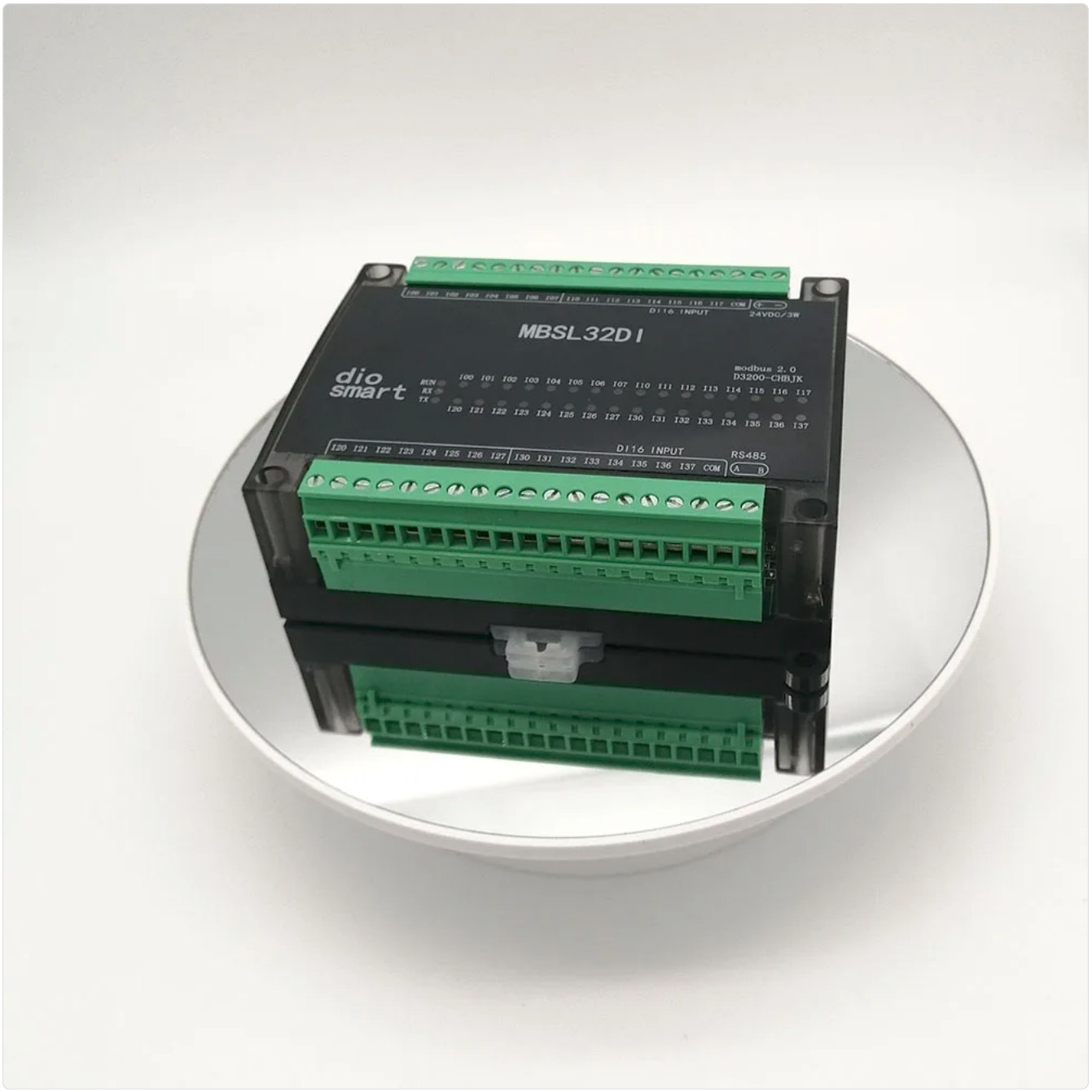


Figure 3.1: OMDAZZ MBSL32DI Module Overview

4. Wiring Instructions

Carefully follow the wiring diagram and terminal definitions below to ensure correct and safe operation of the MBSL32DI module. The module requires a 12V to 30V DC power supply.

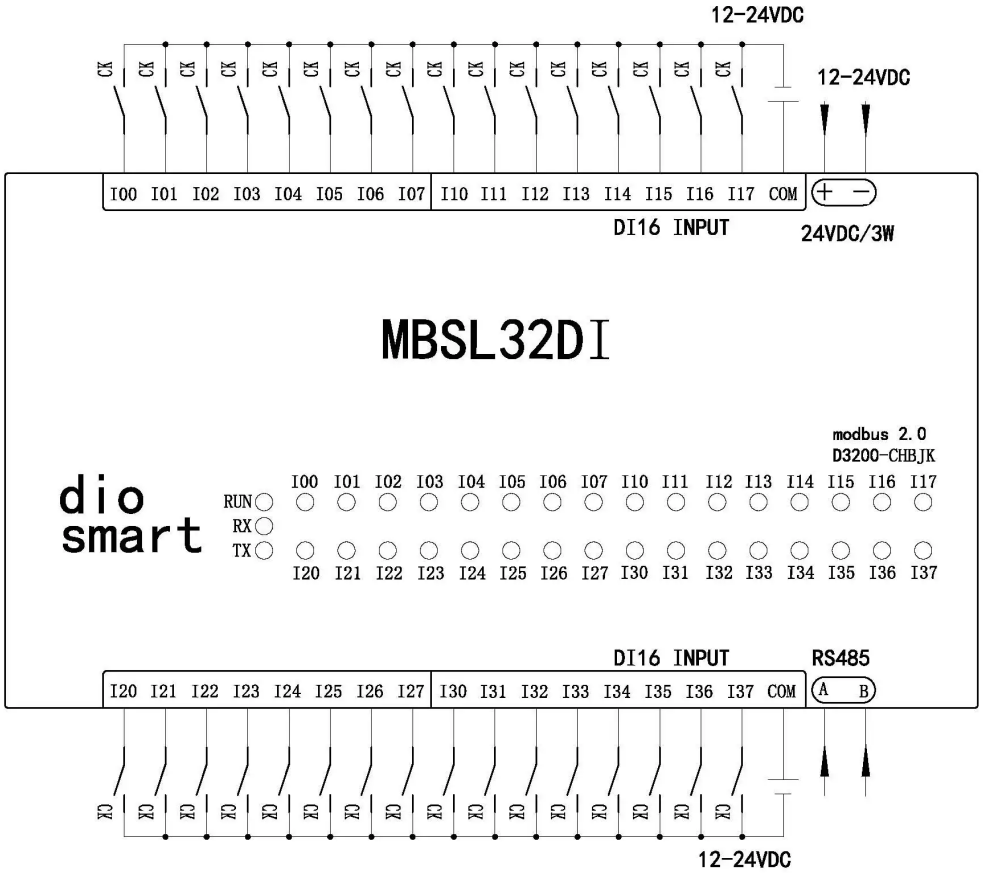


Figure 4.1: MBSL32DI Wiring Diagram

4.1 Terminal Definitions

Terminal name	Description	Terminal name	Description
PW+	External power input positive terminal	485A	RS485 signal A+
PW-	External power input negative terminal	485B	RS485 signal B-
I00	Digital input channel 1	I20	Digital input channel 17
I01	Digital input channel 2	I21	Digital input channel 18
I02	Digital input channel 3	I22	Digital input channel 19
I03	Digital input channel 4	I23	Digital input channel 20
I04	Digital input channel 5	I24	Digital input channel 21
I05	Digital input channel 6	I25	Digital input channel 22
I06	Digital input channel 7	I26	Digital input channel 23
I07	Digital input channel 8	I27	Digital input channel 24
I10	Digital input channel 9	I30	Digital input channel 25
I11	Digital input channel 10	I31	Digital input channel 26

Figure 4.2: MBSL32DI Terminal Definitions

Terminal Name	Description
PW+	External power input positive terminal (12V-30V DC)
PW-	External power input negative terminal (GND)
485A	RS485 signal A+
485B	RS485 signal B-
I00 - I17	Digital input channels 1-16
I20 - I37	Digital input channels 17-32
COM	Common terminal for digital inputs

5. Operating Instructions

The MBSL32DI module operates using the Modbus RTU communication protocol. To configure and operate the module:

1. **Power On:** Connect the module to a 12V to 30V DC power supply via the PW+ and PW- terminals. Observe the RUN LED for proper operation.
2. **RS485 Connection:** Connect the RS485 A+ and B- terminals to your host device (PLC, PC with RS485 converter, etc.).
3. **Digital Inputs:** Connect your digital input signals to the I00-I37 terminals, using the COM terminal as the common reference.
4. **Software Configuration:** Use appropriate Modbus RTU master software to configure the module's address, baud rate, and parity bit. These settings must match the master device's communication parameters.
5. **Data Acquisition:** Once configured, the master device can read the status of the 32 digital input channels via Modbus RTU commands.
6. **Status Indicators:** The RUN, RX (receive), and TX (transmit) LEDs provide

visual feedback on the module's operational and communication status.

6. Maintenance

The MBSL32DI module is designed for robust industrial use and requires minimal maintenance. However, regular checks can ensure its longevity and reliable performance:

- **Environmental Conditions:** Ensure the module operates within the specified temperature (-10 to +60℃) and humidity (35-85% non-condensing) ranges. Avoid exposure to excessive dust, moisture, or corrosive substances.
- **Connections:** Periodically inspect all wiring connections (power, RS485, digital inputs) to ensure they are secure and free from corrosion or damage.
- **Cleaning:** If necessary, gently clean the module's exterior with a soft, dry cloth. Do not use liquid cleaners or solvents.
- **Firmware Updates:** Check the manufacturer's website for any available firmware updates that may improve performance or add features.

7. Troubleshooting

- **Module Not Powering On:**
  - Verify the power supply voltage is within the 12V to 30V DC range.
  - Check the polarity of the power connections (PW+ and PW-).
  - Ensure the power supply is providing sufficient current.
- **No Communication via RS485:**
  - Check the RS485 A+ and B- connections for correct polarity and continuity.
  - Verify that the module address, baud rate, and parity bit settings in

your software match the module's configuration.
    - Ensure the RS485 network is properly terminated if required for your setup.
    - Check the RX and TX LEDs on the module for activity. If no activity, check the master device's communication.
- **Digital Inputs Not Responding:**
    - Ensure the input signals are correctly wired to the Ixx terminals and the common terminal (COM).
    - Verify that the input signal levels are compatible with the module's specifications.
- **Output Disconnection Protection Triggered:**
    - If the module automatically disconnects output, it indicates a loss of communication with the master station for a prolonged period. Re-establish the RS485 communication link and check for network stability.

8. User Tips

While no specific user tips were available from reviews or Q&A, here are some general recommendations for optimal use:

- Always double-check wiring before applying power to prevent damage.
- For long RS485 communication lines, consider using appropriate termination resistors to prevent signal reflections.
- Keep a record of your module's configured address, baud rate, and parity settings for easy reference.
- When integrating with new software or hardware, start with basic communication tests before implementing complex control logic.

9. Warranty and Support

For warranty information, technical support, or service inquiries, please contact your vendor or the manufacturer directly. Ensure you have your product model number (MBSL32DI) and any relevant purchase details available when seeking assistance.

10. Compliance Information

⚠ WARNING: This product can expose you to chemicals including lead, which is known to the State of California to cause cancer and birth defects or other reproductive harm. For more information go to www.P65Warnings.ca.gov.

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